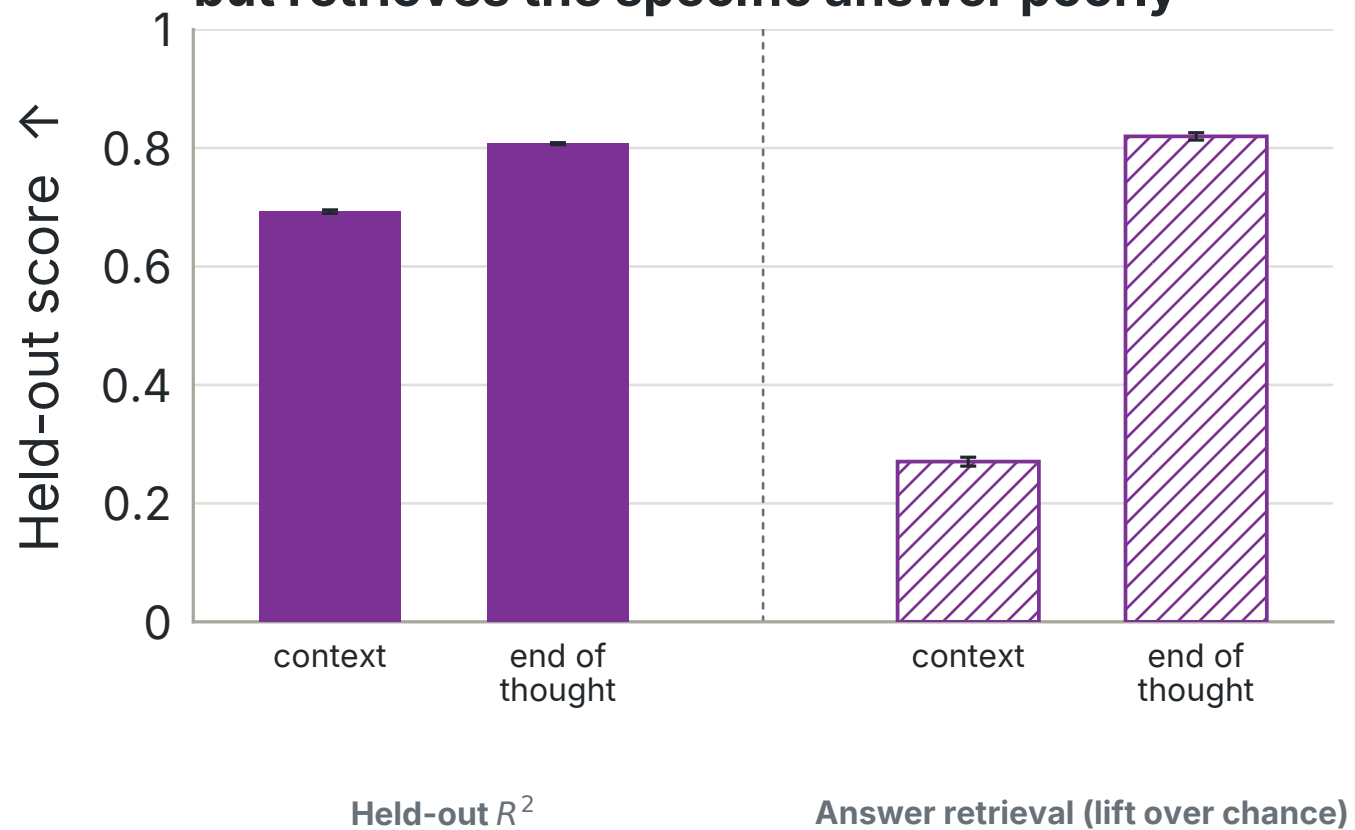


CORPORA

Needs-reasoning corpora only

A · OPENTHINKER3-7B, LAYER 19

The context state predicts the answer state,
but retrieves the specific answer poorly

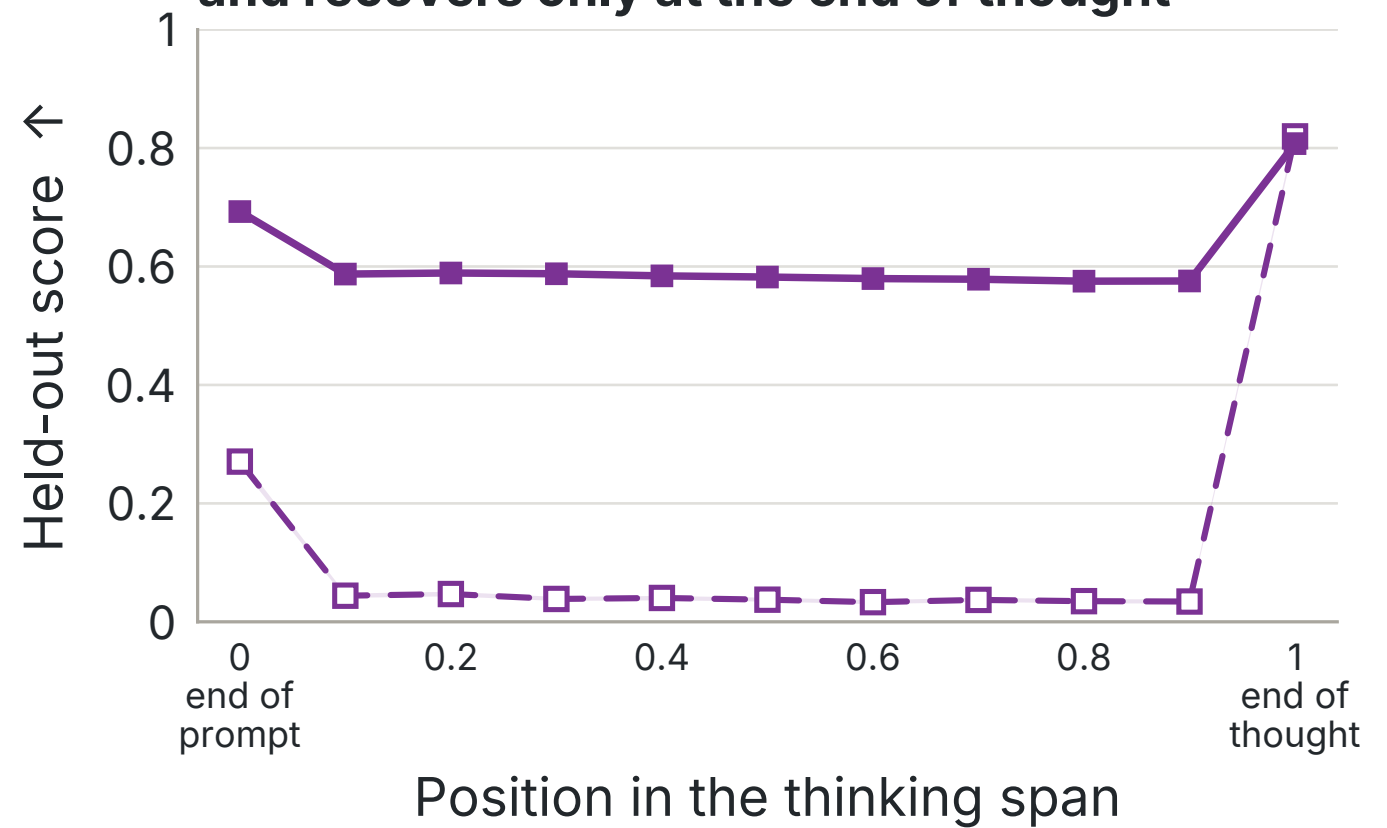


METRIC

Held-out R^2 Answer retrieval (lift over chance)

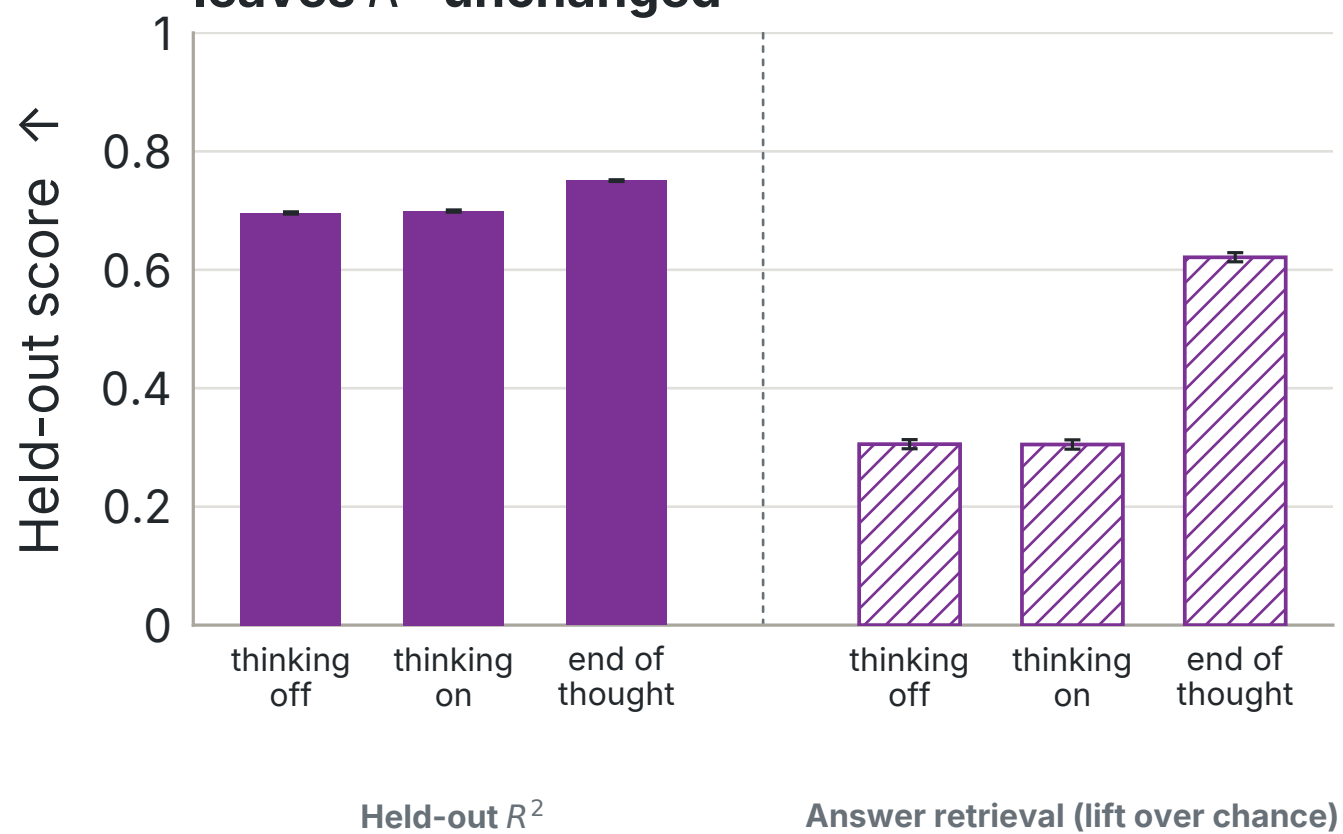
B · OPENTHINKER3-7B, LAYER 19

Predictability dips inside the reasoning trace
and recovers only at the end of thought



C · QWEN3-8B, LAYER 24

Disabling thinking on the same weights
leaves R^2 unchanged



D · QWEN2.5-7B-INSTRUCT (BEFORE) → OPENTHINKER3-7B (AFTER), LAYER 19

The context state before reasoning training
predicts the answer state after it

